

Evaluation of the effect of morphological features of flow paths on solute transport by using fractal dimensions of methylene blue staining pattern

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ABSTRACT

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Fractal dimensions of methylene blue staining patterns in five undisturbed soil columns were used to explain solute transport in five soils. Columns, 15 cm in diameter and 15 cm in height, were taken from each B or BC horizon of a Brown Forest soil (BFS), a Dark Red soil (DRS), a Pseudogley (PSG), a Brown Andosol (ADO) and a Volcanogenous Regosol (VRS). Fractal dimensions of stains were related to Brenner numbers of chloride breakthrough measured under a saturated steady-state condition in each column. The Brenner number was smaller in columns showing the most rapid initial elevation of concentration. The degree of the effect of the macropore was in the order of $DRS > PSG > VRS > BFS > ADO$.

The fractal dimension of the internal structure of staining patterns (D_s), decreased with an increase in depth, ranging from 0.59 to 2.0. A value less than 1 was obtained for stains found in parts of cracks and in root channels in the deeper layers of DRS and PSG columns, while a value greater than 1.7 was obtained for large stains found throughout the ADO column. On the other hand, the fractal dimension of the perimeter of stains (D_{pe}) was estimated to be almost the same through each column; the depth average ranged from 1.30 for ADO to 1.64 for VRS. Smaller values of D_{pe} indicated that the stained parts had smoother boundaries.

An empirical equation, which explains the Brenner number (B) for chloride breakthrough as a function of depth averages of both fractal dimensions and macroporosity (θ_m), was obtained as follows:

$$B = a\theta_m^{-\bar{D}_s} + b\bar{D}_{pe} + c$$

When $\bar{D}_s$ was taken to be the average value of D_s obtained in the lower half of each column, the relation regressed with a correlation coefficient of 0.9995. This equation suggests that the macropore flow contributes significantly to miscible displacement and that this contribution depends on the internal structure of the stains and on the smoothness of their boundaries.

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INTRODUCTION

Flow of water and solutes is affected by the arrangement and continuity of pores. When present, movement of water and solutes occurs preferentially through continuous macropores. The effect of macropores has been identified in terms of a rapid rise of the concentration of chloride or tritiated water in the early stage of breakthrough and a model for this type of breakthrough has been proposed (Bouma et al., 1983; White et al., 1984; Hatano et al., 1985). Dye tests have also been conducted to observe directly the actual flow paths (e.g. Bouma and Dekker, 1978; Hatano et al., 1983). Effective flow paths are present only in part of the macropores. A model for predicting bypass flow used such observations (Hoogmoed and Bouma, 1980; Hatano et al., 1984). Furthermore, Bouma et al. (1979) have attempted to explain numerically the relationship between saturated hydraulic conductivity (K_{sat}) and soil pore geometry, which they measured by infiltrating a methylene blue solution as a tracer in saturated, undisturbed soil columns. They found that K_{sat} was governed by small pore necks in the flow system, and that the volumetric percentage of stained pores was considerably less than 1%.

These studies suggest that soil morphology is useful in evaluating irregular transport processes in soils with macropores. However, dye patterns have only been described in fragmented general terms and there is a clear need to obtain comprehensive parameters. Fractal theory, which has been proposed by Mandelbrot (1967), may provide the means to do so.

Fractal theory provides a means to measure irregular or rough structures by a property called the fractal dimension. For example, Lovejoy (1982) reported that use of radar disclosed that the value of the fractal dimension of cloud perimeters, whose areas ranged from 1 to $1.2 \times 10^6 \text{ km}^2$, was 1.35. This result indicated that the roughness of the perimeter of the clouds was similar for clouds of different cross-sectional area. Morse et al. (1985) showed that the value of fractal dimensions of outlines of various plants, shown on photographs, ranged from 1.28 to 1.79, and that plants with a higher fractal dimension had a more complex shape. Hatano and Sakuma (1990) reported a linear relation between fractal dimensions of distribution patterns of macropores and plant roots. Furthermore, Tyler and Wheatcraft (1990) designed a fractal model to predict water retention curves based on a Sierpinski carpet. The model theoretically explained the empirical water retention model of Campbell (1974).

The purposes of this paper are to evaluate (1) the geometry of flow paths in different soils, as characterized by methylene blue, by using fractal dimensions; and (2) the effect of soil morphology, thus characterized, on solute transport by investigating the relationship between the Brenner number for chloride breakthrough and the fractal dimensions of dye patterns.

EXPERIMENTAL PROCEDURE

Soil used

Five undisturbed soil samples were taken in steel cylinders, 15.5 cm in diameter and 15 cm in height, from the B or BC horizons of five pedons: an Acid Brown Forest soil (BFS), a Dark Red soil (DRS), a Pseudogley (PSG), a Brown Andosol derived from Eniwa tephra (ADO), and a Volcanogenous Regosol derived from Tarumae-d tephra (VRS) (Hokkaido Soil Classification Committee, 1979), located respectively at Aoyama, Tokotan, Shibetsu, Tsukisamu and Bibi in Hokkaido, Japan. These soils are classified as Dystric Cambisol, Humic Acrisol, Dystric Gleysol, Ochric Andosol and Andic Regosol, respectively (FAO, 1988). Sampling was conducted by slowly pressing the cylinders into the soils to minimize disturbance.

Texture and the characteristics of the macro-structures of these samples are presented in Table 1. BFS and DRS were characterized by relatively well developed fine subangular blocky or medium blocky peds. PSG had medium prismatic and blocky peds with a few fine planes. ADO was characterized by moderately developed spherical peds of about 15 mm in diameter buried within loose soil materials. VRS consisted of porous pumice stones, 8–15 mm in diameter.

TABLE 1

Texture and macro-structure of soils and one pore volume (1PV), macropore volume (MPV), and saturated hydraulic conductivity (K_{sat}) measured in soil columns of 2830 cm³.

Characteristic	Soil				
	BFS	DRS	PSG	ADO	VRS
Texture	LiC	LiC	SiC	LiC	pumice
Structure					
Aggregates	Sb	Sb, B	Pr, B	Sph	Sph
Size	F	F, M	M, M	M	M
Grade	S	M, S	S, S	M	S
Voids	Vu, Pn	Chn, Pn	Pn	Cxp	Pn
1PV (cm ³)	1616	1619	1670	1947	2293
MPV (cm ³)	133	102	214	178	647
K_{sat} (m/s)	2.19×10^{-5}	1.21×10^{-5}	8.30×10^{-5}	7.30×10^{-5}	1.20×10^{-2}

Aggregates: Sb = Subangular blocky; B = Blocky; Pr = Prismatic; Sph = Spherical.

Size: F = Fine; M = Medium.

Grade: S = Strongly developed; M = Moderately developed.

Voids: Vu = Vughs; Chn = Channels; Pn = Planes; Cxp = Complex packing.

Breakthrough experiment

After the soil columns had been transported to the laboratory, the bottom of each column was covered with filter paper and placed in 5 cm deep distilled water for about 1 week to allow water-saturation of the sample. The saturated hydraulic conductivity for each column was then measured by the falling head method (Klute and Dirksen, 1986). The column was placed on a funnel and was further saturated with distilled water drawn up slowly from the bottom by a microtube-pump. After 1 mm of water was ponded on the surface of the column, a 0.01N methylene blue solution ($C_{16}H_{18}N_3SCl$) was poured onto the surface with a mariotte device to maintain shallow ponding; at the same time percolation was started. When methylene blue dissolves in water, it dissociates into Cl^- and $C_{16}H_{18}N_3S^+$ (MB^+), the latter acting as a blue dye. The replacement of water in the column by methylene blue solution was continued by a regulating microtube-pump under a steady-state condition at a flow rate of 1.06 cm/h. This flow rate is slower than the saturated hydraulic conductivities of the soil columns, which are presented in Table 1.

Every 20 ml of percolating liquid was collected by a fraction collector until the cumulative amount of percolation reached 4 liters. After replacement of the water, the volume of gravitational drainage was measured to define this volume as the volume of macropores in the soil column. The column was then cut horizontally into 9 layers with a thickness of 0.5 to 2 cm. Water content in each layer was measured and one pore volume (1PV) in the soil column was obtained as the sum of the volume of macropores and the volume of water contained in the soil layers.

The concentration of MB^+ in the percolating solution was directly measured by colorimeter at 710 nm after suitable dilution of the solution. The concentration of chloride was measured by ion-chromatography after the blue color of MB^+ in the solution was removed by utilizing the characteristic ability of kaolinite powder to absorb MB^+ . To obtain a colorless solution, we took 15 ml of the original solution with 0.5 g of kaolinite powder and shook it vigorously for about 1 min; it was then filtered through a membrane filter with a mesh size of 0.2 μm .

Characterization of chloride breakthrough

The chloride breakthrough curves were characterized by Brenner number (B) which is defined as $B = U \cdot L / Dis$ where U is the pore water velocity, L is the column length and Dis is the dispersion coefficient. The Brenner number was calculated according to the empirical law introduced by Rose and Pas-sioura (1971); that is, if the log-probability plots between the effluent volume expressed in terms of pore volume units (pv) and the relative concentration (C/C_0) fall onto a straight line, the following equation can be obtained:

$$\text{inverf}(2 \cdot C/C_0 - 1) = S \ln PV + c \quad (1)$$

where *inverf* is the inverse error function, and *S* and *c* are regression coefficients. By using the value of *S*, *B* can be calculated from the following equation:

$$\log B = 0.1139(\log S)^3 - 0.3504(\log S)^2 + 2.3623(\log S) + 0.4732 \quad (2)$$

Observation of the area stained by methylene blue

Blue stains were observed on horizontal cuts through the column. The image was scanned in a rectangle of 512 by 480 pixels by a CCD color video camera after fixing diagonal lines of the rectangle of pixels to the diameter of the soil column. The width of a pixel was consequently calibrated to 0.0221 cm. The signal data of the image were passed to an image analyzer, NEXUS 6400, controlled with a 16-bit personal computer, NEC-PC9801VM. We were then able to easily isolate the blue stains in the image analyzer on the basis of their grey level by regulating the exposure of the camera. The number, area and perimeter of stains were measured with the image analyzer.

If a stain consisted of only a pixel, however, the image analyzer measured both the area and perimeter as the same value. This is not a realistic value and was therefore omitted.

Estimation of fractal dimensions

Two kinds of fractal dimension were estimated. One (D_{pe}) is called the fractal dimension of the perimeter. D_{pe} characterizes the smoothness of the boundary line of stained parts, and the value is estimated in the range of 1 (for the smoothest line) to 2. In practical terms, it is easy to get the D_{pe} value from the double logarithmic relationship between perimeter (*Pe*) and area (*A*). The relation can be written as:

$$\log A = (2/D_{pe}) \log Pe + c \quad (3)$$

where *c* is a constant. If the double logarithmic plots for the perimeters and the areas of stains are correlated to a straight line, D_{pe} can be estimated from its slope. This method is the same as Lovejoy (1982) used to measure D_{pe} values of clouds. The values of *A* and *Pe* of particular stains, which were cut off by the sampling frame of the image analyzer, will result in an error when estimating D_{pe} . The image analyzer used can, however, eliminate these data. This elimination, however, did not affect the estimation of D_{pe} in most cross-sections of samples other than the top three cross sections in ADO which were almost entirely stained. In these three cross sections we were not able to ob-

tain the D_{pe} values because the number of stains was not enough for the regression of eq. (3).

Another fractal dimension (D_s) is related to the basic shape of an object, be it for an example a dot, line, or plane. This procedure is similar to the one presented by Morse et al. (1985). The value is estimated in the range of 0 to 2. D_s is measured by dividing a horizontal stained cross-section into squares of width r (the smallest value of r corresponding to the width of an original pixel), and then counting the number of partially stained squares, $N(r)$. A series of $N(r)$ values is obtained by changing the r value in integral multiples of the width of an original pixel. If the double logarithmic plots of $N(r)$ against r are correlated with straight lines, D_s is estimated from the slope of the lines, as follows:

$$\log N(r) = -D_s \cdot \log r + c \quad (4)$$

where c is a constant. For example, as shown in Fig. 1, the $\log N(r) - \log r$ plots for dots scattered at nodes of a grid, with section width h , size R by R , fall into two straight lines which cross each other at r equal to h . In the range $r < h$, $N(r)$ is maintained as a number of dots, $(R/h)^2$, and D_s is obtained as 0. On the other hand, in the range $r > h$, $N(r)$ changes as $(R/r)^2$ and then D_s becomes 2. Furthermore, the distribution pattern of parallel lines at h intervals shows D_s in the range $r < h$ is obtained as 1, while D_s in the range $r > h$ becomes 2. D_s for a plane is obtained as 2 in the whole range of r . These examples allow us to recognize D_s in the first range of $r < h$ as a fractal dimension of the structure of distribution patterns.

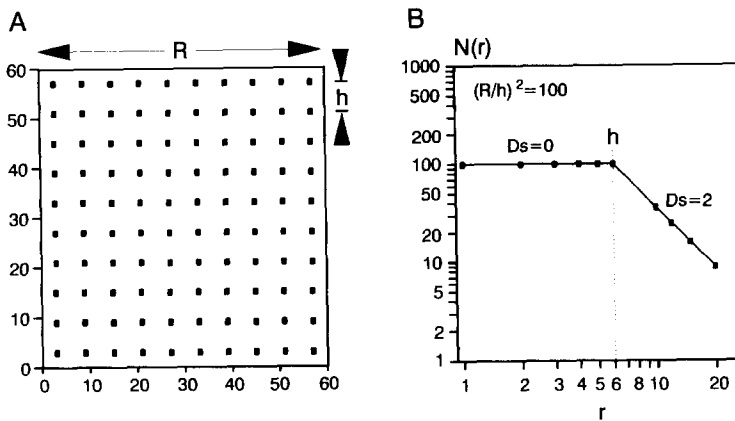


Fig. 1. Double logarithmic relationship between the width of square as a divider (r) and the number of squares containing the object ($N(r)$) for dots scattered at nodes of a grid, with section width h , size R by R .

RESULTS AND DISCUSSION

Breakthrough curves of chloride and MB⁺

Figure 2 shows the breakthrough curves of chloride and MB⁺ for each column. One pore volume and macropore volume are presented in Table 1 for each column. BFS showed a linear and gradual elevation of chloride concentration but a considerably slower breakthrough of MB⁺. DRS and PSG were characterized by rapid elevation of C/C_0 for both chloride and MB⁺ during the initial stage of breakthrough and a significantly long tailing away during the later stage. Especially in DRS, there was a very rapid elevation of both chloride and MB⁺; that is, when effluent reached a macropore volume of 102 ml (Table 1), C/C_0 for chloride and MB⁺ showed the very large values of 0.45 and 0.21, respectively. The shape of the breakthrough curve of chloride for VRS was intermediate between that for PSG and BFS, but the breakthrough of MB⁺ was significantly faster than in PSG, and it was very similar to that of chloride. ADO showed slower breakthrough of chloride than did BFS, but a faster breakthrough of MB⁺ than in BFS.

The log-probability plots for all the chloride breakthrough curves regressed linearly with correlation coefficient values greater than 0.99 (Fig. 2). The Brenner numbers are presented in Table 2. These reveal that smaller values were found in samples which showed a more rapid elevation of the C/C_0 during the initial stage of breakthrough.

The rapid elevation of the concentration during the early stages of chloride breakthrough can be ascribed to the effect of macropores (White et al., 1984). This includes preferential flow in macropores and a stagnant water regime of micropores which is being bypassed. When judged by the shape of chloride breakthrough curves and Brenner number, the effect of macropores on flow seems to decrease in the following order: DRS > PSG > VRS > BFS > ADO.

Fractal dimension of internal structure of staining pattern.

The vertical distribution of D_s through each soil column is shown in Fig. 3. In BFS, DRS, and PSG the D_s values decreased significantly throughout the column, starting from 1.7 at a depth of 0.5 cm. As shown in Fig. 4A, BFS was characterized by many fine planar stains from the upper to the lower parts of the column. D_s values were greater than 1.15 throughout the column. Figures 4B and 4C show that in DRS and PSG the stain patterns in the upper layers of the columns were similar to those of BFS, but in the deeper layers stains were only found in root channels and in parts of the cracks around coarse prismatic peds. In DRS, the D_s value at a depth of 11.5 cm was < 1. At a depth of 13 cm the extremely small value of 0.57 was found. This value, an intermediate value between the dimensions of dots and lines, indicates that the

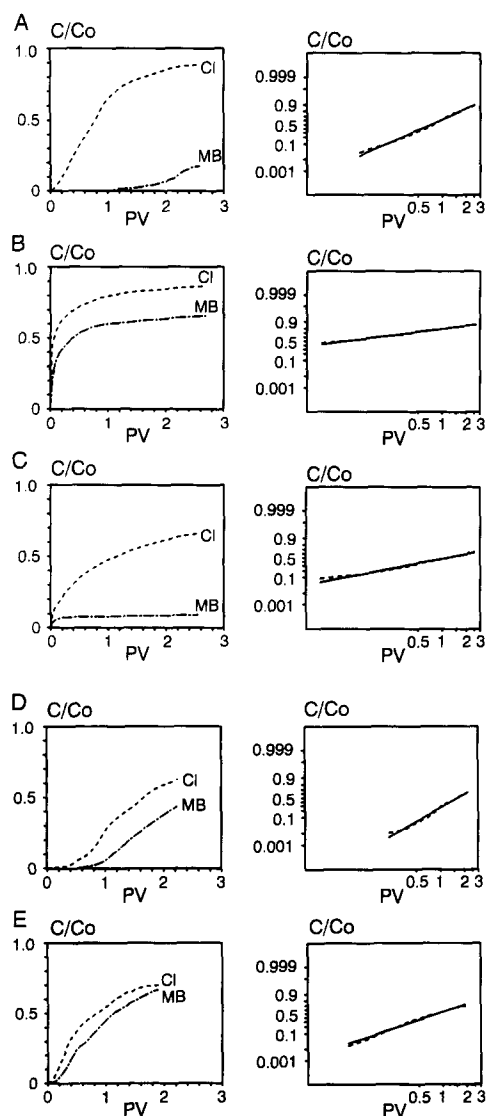


Fig. 2. Breakthrough curves of chloride (*Cl*) and methylene blue dye (*MB*) in each column and log-probability plots for chloride breakthrough curves. (A) BFS, (B) DRS, (C) PSG, (D) ADO, (E) VRS.

contribution of planar voids to flow paths has been lost. Furthermore, PSG, which showed a D_s value of 0.96 at a depth of 13 cm, had intermediate values, somewhere between DRS and BFS. ADO had higher D_s values, which ranged from almost 2 as deep as 3.5 cm to 1.7 at a depth of 11.5 cm. This supports numerically the result of the visual observation that the flow regime is almost two-dimensional and occurs through both macropores and groundmass (Fig.

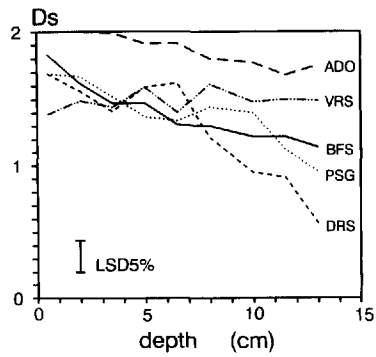


Fig. 3. Vertical distribution of fractal dimensions for internal structure of methylene blue stains (D_s) in soil columns.

4D). VRS had a relatively small variance with an average D_s value of 1.48 which reflects the net-like staining pattern (Fig. 4E).

Fractal dimensions of the perimeter of stains

Figure 5 shows the vertical distribution of D_{pe} through each column. The relations of $\log A$ – $\log Pe$ for all samples were linear, with high correlation coefficients of over 0.98 (data not shown). For a given sample, the values of D_{pe} remained virtually constant with depth. The average D_{pe} value for each sample was: 1.58 for the planar stains between subangular blocky peds in BFS;

TABLE 2

Brenner number (B), macroporosity (θ_m), averaged fractal dimension for perimeter through column ($\bar{D}_{pe}$), and averaged fractal dimensions for structure through column ($\bar{D}_{sw}$) and in lower half of column ($\bar{D}_{sl}$), and the regression equations for B

Parameter	Soil				
	BFS	DRS	PSG	ADO	VRS
B	1.325	0.047	0.147	2.393	0.650
θ_m	0.046	0.036	0.076	0.062	0.229
$\bar{D}_{pe}$	1.58	1.43	1.45	1.30	1.64
$\bar{D}_{sw}$	1.36	1.21	1.35	1.85	1.49
$\bar{D}_{sl}$	1.22	0.91	1.23	1.75	1.51

$$B=0.023 \theta_m^{-\bar{D}_{sw}}+5.755 \bar{D}_{pe}-9.137, \quad R=0.973$$

$$B=0.028 \theta_m^{-\bar{D}_{sl}}+4.673 \bar{D}_{pe}-7.249, \quad R=0.9995^{**}$$

******significant at the 1% level.

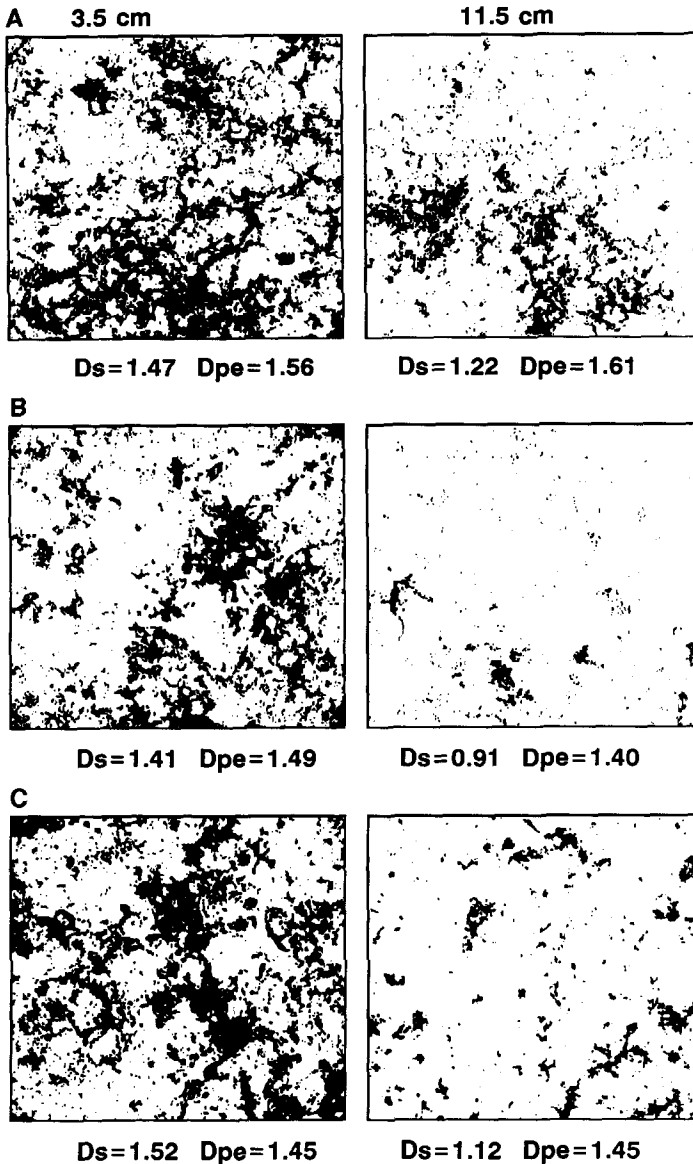


Fig. 4. Images of the distribution of stains at the depths of 3.5 cm and 11.5 cm in soil columns. (A) BFS, (B) DRS, (C) PSG, (D) ADO, (E) VRS.

1.43 for the stains found on the walls of root tubes and parts of the planes between blocky peds in DRS; 1.45 for the stains found in parts of cracks between the coarse prisms in PSG; 1.29 for two-dimensional extensive stains in ADO which have a boundary at the surface of spherical peds; 1.63 for the stains found in the voids around pumice stones in VRS. A smaller value of D_{pe} indicates that the soil has a smoother boundary of the stained parts. This

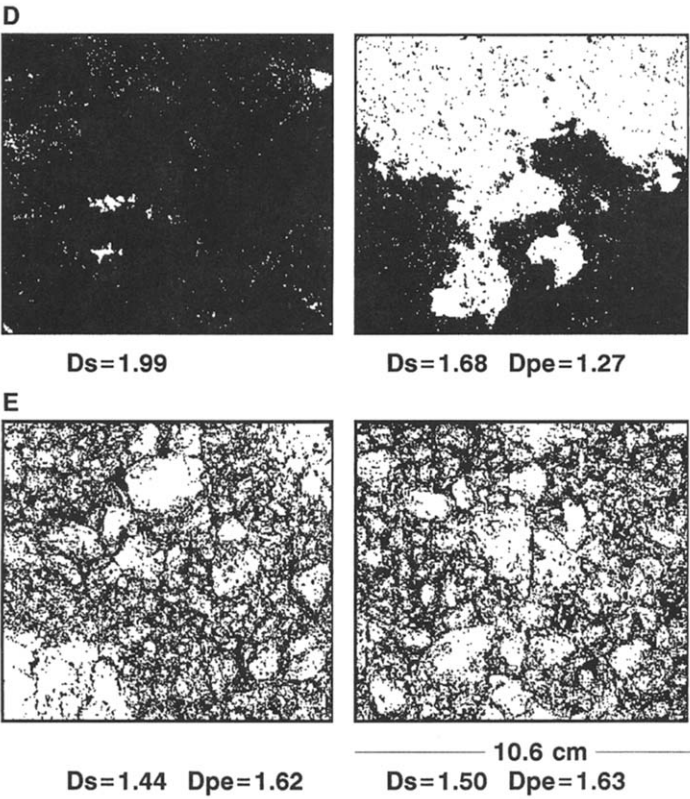


Fig. 4. Continued.

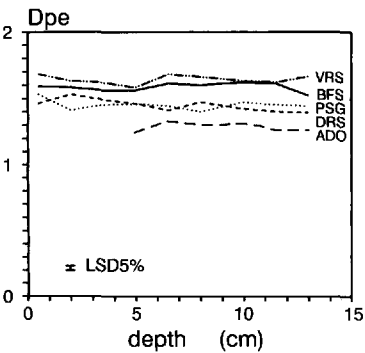


Fig. 5. Vertical distribution of fractal dimensions for perimeter of methylene blue stains (D_{pe}) in soil columns.

corresponds to our experiences based on observations; the surface of porous pumice stones is rough and irregular, while well developed subangular blocky peds have more rough and irregular surfaces than large prismatic or blocky peds. Furthermore, we know empirically that root channels have relatively smooth walls.

Fractal dimensions as a factor determining flow condition

A Brenner number reflects the complex combined effect of water velocity and dispersion coefficient in soil. As an attempt to estimate the Brenner number, we examined the relation of macroporosity and two fractal dimensions of stains to the Brenner number by using multiple regression. Values and regression equations are presented in Table 2. The basic equation is:

$$B = a \cdot \theta_m^{-\bar{D}_s} + b \cdot \bar{D}_{pe} + c \quad (6)$$

where B is the Brenner number, θ_m is macroporosity, and $\bar{D}_s$ and $\bar{D}_{pe}$ are the average values over depth of the fractal dimensions for structure and perimeter, respectively. Since the Brenner number represents an averaged value during chloride breakthrough, also average values for macroporosity and the two fractal dimensions should be used. The values of macroporosity were obtained by dividing the volume of macropores as presented in Table 1 by the total column volume. When the average value of D_s through the column ($\bar{D}_{sw}$) was used for the regression analysis, the following equation was obtained:

$$B = 0.023 \theta_m^{-\bar{D}_{sw}} + 5.755 \bar{D}_{pe} - 9.137 \quad (6')$$

with a correlation coefficient of 0.973. But this equation was not significant even at the 5% level. When the average value of D_s obtained in the lower half of a column ($\bar{D}_{sl}$) was used, the following equation was obtained:

$$B = 0.028 \theta_m^{-\bar{D}_{sl}} + 4.673 \bar{D}_{pe} - 7.249 \quad (6'')$$

with a correlation coefficient of 0.9995. This equation and regression coefficients were significant at 1% level.

Although eq. (6) is empirical, it provides valuable clues. The reciprocal of θ_m indicates the number of displacements of macropore-solution by percolating solution when total effluent has reached the volume of the column. Although such displacement has been evaluated on the basis of a two-dimensional flux, the first term of the right hand side of eq. (6) suggests that the flow has a certain dimension as it includes D_s . This dimension depends on the type and shape of the macropores. This may explain the result of Pas-

sioura and Rose (1971) who related dispersion coefficients in aggregated porous media to the first to second power of the product of flow velocity and aggregate size. Furthermore, the second term of the right hand side, which includes D_{pe} , indicates that the fractal dimension of the stain-perimeters affects solute movement independently.

CONCLUSIONS

(1) A method to evaluate flow paths in soils using methylene blue solution as a tracer, as demonstrated in this study, proved to be effective. The two fractal dimensions described in this paper appear to be promising parameters to independently characterize flow patterns with the objective to predict miscible displacement.

(2) Macropore flow contributes significantly to patterns of miscible displacement and is characteristically different in different soils, depending on the type of macropores present and the smoothness of their walls.

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